

DR GEORGE P. JENKINSON

Postdoctoral Researcher · ARIA Robot Dexterity programme · Department of Mechanical Engineering, University of Bath

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publications

Robotics researcher exploring mathematically driven robotic mechanisms to better understand ourselves: tactile sensing, soft robotics, dexterous manipulation, and assistive technology, with first-author publications spanning swarm robotics, soft robotic sensors, medical and assistive robotics, machine vision, qualitative public engagement, and generative AI.

EMPLOYMENT

Postdoctoral Researcher — **University of Bath**, Department of Mechanical Engineering · Feb 2026 – present
ARIA Robot Dexterity programme. Wave-based control for in-hand manipulation; soft actuator classification and manipulation frameworks.

Senior Research Associate — **University of Bristol**, School of Engineering Mathematics and Technology · Oct 2025 – Feb 2026

Braille-pen project (EPSRC Exploratory Award, Co-I): developing the Braille-tip prototype into a marketable product and investigating market opportunities, with Bristol Braille Technology.

Research Associate — **Centre For Print Research, UWE Bristol** · Mar 2024 – Feb 2026

AI-driven accessibility for heritage collections: machine-vision interpretation and tactile reproduction of embroidered artworks for visually impaired audiences; industry projects with SS Great Britain Museum, British Motor Museum, and The Building Archives.

Teaching Associate — **University of Bristol**, School of Engineering Mathematics and Technology · Dec 2024 – Sep 2025

Supervised 17 individual MSc projects across Robotics disciplines and Data Science, including three data-science industry partnerships; coordinated inter-institutional marking for the Bristol/UWE Robotics MSc; created targeted revision material for preliminary engineering-maths courses.

Graduate Teacher — **University of Bristol** · Sep 2020 – Feb 2022

Production Technician — **Open Bionics** (mechatronic prosthetics), Bristol · Aug 2018 – Apr 2019

Fabrication and testing of sEMG-controlled prosthetic arms; bio-interfacing devices, rapid prototyping, assistive-technology standards.

Private Tutor, Amsterdam, NL · Sep 2016 – Apr 2019

Wrote and delivered a 20-unit course over 2 years, accredited by Clonlara School as equivalent to AS-level Physics, Maths, Chemistry, and Biology.

Research Intern — **AMOLF**, Soft Biological Matter Group, Amsterdam, NL · Jan – Dec 2017

EDUCATION

PhD Mechanical Engineering — Bristol Robotics Laboratory, University of Bristol · Oct 2019 – Feb 2024

Thesis: *Soft Variable Stiffness Tactile Sensors for Clinical Examinations*. Supervisors: Dr Antonia Tzemanaki, Prof Andrew Conn. Funded by the EPSRC Doctoral Training Partnership (EP/R513179/1).

MSc Robotics, Distinction — University of Bristol & University of the West of England · Sep 2018 – Aug 2019

Thesis: *A Bio-Inspired Mechanism for Dynamic Role Allocation in Robotic Swarms: Self-Organised Personality Dynamics for Swarm-Size Sensitive Risk-Taking*

BSc Liberal Arts & Sciences, Cum Laude — University College Amsterdam · Feb 2015 – Feb 2018

Major in Physics. Thesis: *A Novel Holistic Analysis Method for the Mechanical Behaviour of Giant Unilamellar Vesicles* — Thesis of Distinction Award (Soft Matter Biophysics)

PUBLICATIONS

* denotes equal contribution

Journal articles

- Jenkinson GP**, Conn AT, Tzemanaki A. Coil-Tac: Coiled Capacitor Mechanism with Liquid Metal for Tactile Sensing. *IEEE Robotics and Automation Letters*, 10(5), 5194–5200, May 2025. [doi:10.1109/LRA.2025.3558655](https://doi.org/10.1109/LRA.2025.3558655) · [video](#)
- Wharton P*, You TL*, **Jenkinson GP**, Diteesawat RS, Le NH, Hall EC, Garrad M, Conn AT, Rossiter J. Tetraflex: A Multigait Soft Robot for Object Transportation in Confined Environments. *IEEE Robotics and Automation Letters*,

- 8(8), 5007–5014, 2023. [doi:10.1109/LRA.2023.3290409](https://doi.org/10.1109/LRA.2023.3290409) · [video](#)
3. **Jenkinson GP***, Houghton N*, van Zalk N, Waller J, Bello F, Tzemanaki A. Acceptability of Automated Robotic Clinical Breast Examination: Survey Study. *Journal of Participatory Medicine*, 15, e42704, 2023. [doi:10.2196/42704](https://doi.org/10.2196/42704)
 4. **Jenkinson GP**, Conn AT, Tzemanaki A. ESPRESS.O: Eustachian Tube-Inspired Tactile Sensor Exploiting Pneumatics for Range Extension and Sensitivity Tuning. *MDPI Sensors*, 23, 567, 2023. [doi:10.3390/s23020567](https://doi.org/10.3390/s23020567) · [video](#)
 5. Bewley J, **Jenkinson GP**, Tzemanaki A. Optical-Tactile Sensor for Lump Detection Using Pneumatic Control. *Frontiers in Robotics and AI*, 8, 1–12, 2021. [doi:10.3389/frobt.2021.672315](https://doi.org/10.3389/frobt.2021.672315) · [video](#)
 6. **Jenkinson GP**. Newton the Natural Philosopher: Does Newtonian Physics carry Newton's God? *AUC Undergraduate Journal of Liberal Arts & Sciences*, 56, 2016.

Conference papers

1. **Jenkinson GP**, Aure Calvet X. Digitally Reintegrating Losses in Heritage Embroidery with Model-Independent Stitch Priors and Surface-Normal Verification. *23rd EUROGRAPHICS Workshop on Graphics and Cultural Heritage (GCH 2026)*, 2026. [accepted]
2. **Jenkinson GP**, Samaroudi M, Aure Calvet X. Tactile Embroidery Reproduction Exploiting Machine Vision for Visually Impaired Engagement. *Digital Heritage 2025*, pp. 1–10, Eurographics, 2025. [doi:10.2312/dh.20253349](https://doi.org/10.2312/dh.20253349)
3. **Jenkinson GP**, Aure Calvet X. Matching the Stitch: Autonomous Classification of Embroidery Techniques for Replication. *Textile Conservation Conference @ 50*, Glasgow, UK, Jun 2025.
4. Aure Calvet X, **Jenkinson GP**. Beyond the display case: New methods for documenting and accessing historic textile collections. *Textile Conservation Conference @ 50*, Glasgow, UK, Jun 2025.
5. **Jenkinson GP**, Conn AT, Tzemanaki A. Braille-tip: Structured Small-Footprint Tactile Sensor for High Acuity Dynamic Tactile Tasks. *10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)*, Heidelberg, Germany, Sep 2024. [doi:10.1109/BioRob60516.2024.10719951](https://doi.org/10.1109/BioRob60516.2024.10719951) · [video](#)
6. **Jenkinson GP**, Tiemann K, Papathanasiou A, Bewley J, Conn AT, Tzemanaki A. A robotic Radial palpation mechanism for breast examination (IRIS). *32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, Busan, Republic of Korea, Aug 2023. [doi:10.1109/RO-MAN57019.2023.10309324](https://doi.org/10.1109/RO-MAN57019.2023.10309324) · [video](#)
7. **Jenkinson GP**, Conn AT, Tzemanaki A. A Pressure Controlled Membrane Mechanism for Optimising Haptic Sensing. *Towards Autonomous Robotic Systems (TAROS)*, Nottingham, UK, 2020. [doi:10.1007/978-3-030-63486-5_31](https://doi.org/10.1007/978-3-030-63486-5_31) · [video](#)
8. **Jenkinson GP***, Hunt ER*, Wilsher MJG, Dettmann CP, Hauert S. SPIDER: a Bioinspired Swarm Algorithm for Adaptive Risk-Taking. *ALIFE 2020: The 2020 Conference on Artificial Life*, 2020. [doi:10.1162/isal_a_00279](https://doi.org/10.1162/isal_a_00279)

Thesis

- **Jenkinson G**. *Soft Variable Stiffness Tactile Sensors for Clinical Examinations*. PhD thesis, University of Bristol, 2024.

GRANTS & FUNDED PROJECTS

Research council

- **Braille-pen** · EPSRC Exploratory Award · Oct 2025 – Feb 2026 · **Co-I** Developing the Braille-tip prototype into a marketable product and investigating market opportunities. Partner: Bristol Braille Technology CIC.
- **Matching the Stitch** · AHRC · Oct 2024 – Mar 2025 · **Co-I** AI-based workflow to restore embroidered artworks with true-to-original colouring and texture. Partner: Holburne Museum.

Industry

- **Tactile Car Models from Heritage Photos** · British Motor Museum · Oct 2025 · **Co-I** Digital and physical 3D models of cars from archive photographs for tactile interaction.
- **Digitally Enhanced Artefacts** · SS Great Britain Museum · Sep 2025 · **Co-I** Fine-tuning generative AI models to restore lost details for digital museum display.

Studentship

- EPSRC Doctoral Training Partnership studentship (EP/R513179/1), University of Bristol, 2019–2023.

Bids in development

- **Robotics for Affective Touch** · research council · **PI** · with the Universities of Gothenburg, Linköping, and Birmingham Robotic platform combining a parametric wearable with microneurography to investigate the neural response to social/affective touch.
- **ACCESS Music: Inclusive Music Literacy Platform for the Visually Impaired** · Horizon Europe · **Technical Advisor** · with Edinburgh, Cambridge, Glasgow, Oxford, Oslo, York, UCD, and Western Norway University of Applied Sciences Musical notation device with dynamic Braille output and synchronised auditory feedback.

AWARDS

- **Shortlisted** — TakeAIM, Smith Institute Award for Mathematics in Healthcare · Mar 2024
- **Thesis of Distinction** (Soft Matter Biophysics) — University College Amsterdam · Feb 2018

PRESENTATIONS & EVENTS

- **Invited workshop** — Fine-tuning generative image models: commissioned two-day workshop for researchers (LoRA training across architectures, dataset curation and captioning, evaluation, ComfyUI workflows), with a shared GPU training environment, overnight training queue and take-away repository for independent runs · Centre For Print Research, UWE Bristol · Jul 2026
- **Talk** — Coil-Tac: Coiled Capacitor Mechanism with Liquid Metal for Tactile Sensing · IEEE IROS · Oct 2025 · [video](#)
- **Talk** — Tactile Embroidery Reproduction Exploiting Machine Vision for Visually Impaired Engagement · Digital Heritage · Sep 2025
- **Talk** — Matching the Stitch: Autonomous Classification of Embroidery Techniques for Replication · Textile Conservation Conference @ 50 · Jun 2025
- **Talk** — Harnessing Generative AI Models for Heritage Restoration · National Trust Textile Conservation Studio · Feb 2025
- **Workshop host** — Making Exhibition Spaces Accessible for the Visually Impaired · Oct 2024
- **Talk** — Braille-tip: Structured Small-Footprint Tactile Sensor · IEEE BioRob · Sep 2024 · [video](#)
- **Talk** — From Soft Robotics to Tactile Interfaces · CFPR lunchtime seminars · Jul 2024
- **Talk** — A robotic Radial palpation mechanism for breast examination (IRIS) · IEEE RO-MAN · Aug 2023 · [video](#)
- **Invited speaker** — The ARTEMIS project: using robotics and soft sensors to detect early-stage breast cancer · Interdisciplinary Research Symposium on the Prevention, Early Detection and Treatment of Cancer · Jan 2023
- **Poster** — A Soft Tactile Sensor for Tasks Requiring High Spatial Resolution · IEEE IROS · Oct 2022
- **Poster** — Exploiting Mammalian Inner Ear Mechanics for Increased Pressure Sensitivity and Range · IEEE RoboSoft · Apr 2022
- **Track co-chair** — 1 of 4 tracks · BRL Conference, Bristol Robotics Laboratory · Jun 2021
- **Talk** — SPIDER: a Bioinspired Swarm Algorithm for Adaptive Risk-Taking · ALIFE · Jul 2020
- **Invited speaker** — When More is Different: Fundamental Emergence in Entangled Quantum States · International Interdisciplinary Seminar “Differentiating humans from animals and machines” · Jan 2017

MEDIA COVERAGE

Coverage of my research on autonomous robotic breast examination and the Braille-reading assistive pen, including repeat invitations:

- BBC Radio Bristol (2023, 2024) · BBC News (2023, 2024) · ITV News · The Independent · Somewhere on Earth podcast (2023, 2024) · and more — all coverage embedded at geojenks.github.io/media

TEACHING & SUPERVISION

- Taught on BSc and MSc courses at the University of Bristol across mechanical engineering, control, engineering mathematics, and advanced robotics.
- 12+ years of teaching experience across levels, from a registered tutoring business in Amsterdam to MSc supervision at Bristol.
- Supervised 17 individual MSc projects (2024/25) across Robotics disciplines and Data Science, including three industry-partnered projects; ~3 with publication potential.
- Co-supervised 3 individual MSc projects and 1 group project during PhD, 2 leading to publications (Frontiers in Robotics & AI; RO-MAN).
- Coordinated cross-institution marking for the Bristol/UWE Robotics MSc.
- Wrote and delivered a 2-year, 20-unit course accredited by Clonlara School as AS-level equivalent (Physics, Maths, Chemistry, Biology).

SKILLS

Programming: Python, Lisp, MATLAB, C++ (competent); PHP, HTML, C, Java (working knowledge). **AI/ML:** fine-tuning open-source models, diffusion-based image generation, segmentation, LLMs, ComfyUI, TensorFlow, OpenCV, Weka. **Engineering:** SolidWorks, Arduino, 3D printing, CNC milling, laser cutting, electronics/soldering, robotic arms, elastomeric moulding (soft robotics lab), Biohazard Level 1 laboratory experience.

PROFESSIONAL MEMBERSHIP

- IEEE

REFERENCES

Prof Min Pan — Line manager · Professor, Department of Mechanical Engineering, University of Bath

Dr Xavi Aure-Calvet — Line manager, Mar 2024 – Feb 2026 · Research Fellow, Centre For Print Research, UWE Bristol

Prof Andrew Conn — PhD co-supervisor; line manager as Teaching Associate · Associate Professor of Robotics, University of Bristol

Dr Antonia Tzemanaki — PhD supervisor; PI of the Braille-pen project · Associate Professor of Robotics, University of Bristol

Contact details available on request.